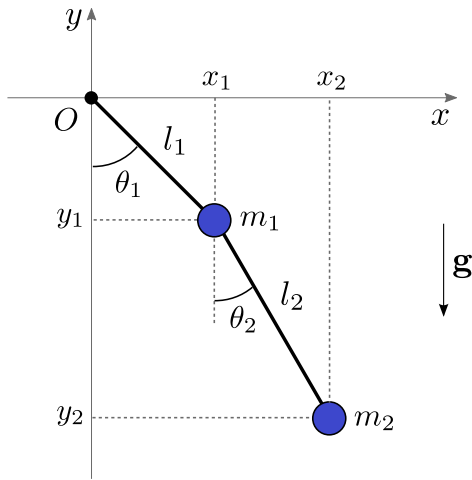


Derivation



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$$y_1 = -l_1 \cos(\theta_1)$$

$$y_2 = -l_1 \cos(\theta_1) - l_2 \cos(\theta_2)$$

$$x_1 = l_1 \sin(\theta_1)$$

$$x_2 = l_1 \sin(\theta_1) + l_2 \cos(\theta_2)$$

$$\dot{y}_1 = l_1 \dot{\theta}_1 \sin(\theta_1)$$

$$\dot{y}_2 = l_1 \dot{\theta}_1 \sin(\theta_1) + l_2 \dot{\theta}_2 \sin(\theta_2)$$

$$\dot{x}_1 = l_1 \dot{\theta}_1 \cos(\theta_1)$$

$$\dot{x}_2 = l_1 \dot{\theta}_1 \cos(\theta_1) + l_2 \dot{\theta}_2 \cos(\theta_2)$$

Finding T and U

$$\begin{aligned} T &= \frac{1}{2} (m_1 \sqrt{\dot{x}_1^2 + \dot{y}_1^2} + m_2 \sqrt{\dot{x}_2^2 + \dot{y}_2^2}) \\ &= \frac{1}{2} m_1 l_1^2 \dot{\theta}_1^2 + \frac{1}{2} m_2 (l_1^2 \dot{\theta}_1^2 + l_2^2 \dot{\theta}_2^2 + 2 l_1 \dot{\theta}_1 l_2 \dot{\theta}_2 \cos(\theta_1 - \theta_2)) \end{aligned}$$

$$U = m_1 g y_1 + m_2 g y_2$$

$$= -(m_1 + m_2) g l_1 \cos(\theta_1) - m_2 g l_2 \cos(\theta_2)$$

$$L = T - U = \frac{1}{2} m_1 l_1^2 \dot{\theta}_1^2 + \frac{1}{2} m_2 (l_1^2 \dot{\theta}_1^2 + l_2^2 \dot{\theta}_2^2 + 2 l_1 \dot{\theta}_1 l_2 \dot{\theta}_2 \cos(\theta_1 - \theta_2)) + (m_1 + m_2) g l_1 \cos(\theta_1) + m_2 g l_2 \cos(\theta_2)$$

Using the Euler-Lagrange equations

$$\frac{d}{dt} \left(\frac{\partial L}{\partial \dot{\theta}_i} \right) - \frac{\partial L}{\partial \theta_i} = 0 \text{ for } i \in 1, 2$$

After some tedious manipulation,

$$l_1 \ddot{\theta}_1 (m_1 + m_2) + l_2 m_2 \ddot{\theta}_2 \cos(\theta_1 - \theta_2) + l_2 m_2 \dot{\theta}_2^2 \sin(\theta_1 - \theta_2) + (m_1 + m_2) g \sin(\theta_1) = 0$$

$$l_2 \ddot{\theta}_2 + l_1 \ddot{\theta}_1 \cos(\theta_1 - \theta_2) - l_1 \dot{\theta}_1^2 \sin(\theta_1 - \theta_2) + g \sin(\theta_2) = 0$$

Shifting all the second-derivatives to one side yields:

$$\ddot{\theta}_1 + \alpha_1 \ddot{\theta}_2 = f_1(\theta_1, \theta_2, \dot{\theta}_1, \dot{\theta}_2)$$

$$\ddot{\theta}_2 + \alpha_2 \ddot{\theta}_1 = f_2(\theta_1, \theta_2, \dot{\theta}_1, \dot{\theta}_2)$$

where

$$\alpha_1 = \frac{l_2 m_2}{l_1 (m_1 + m_2)} \cos(\theta_1 - \theta_2)$$

$$f_1 = -\frac{l_2 m_2 \dot{\theta}_2^2}{l_1 (m_1 + m_2)} \sin(\theta_1 - \theta_2) - \frac{g}{l_1} \sin(\theta_1)$$

$$\alpha_2 = \frac{l_1}{l_2} \cos(\theta_1 - \theta_2)$$

$$f_2 = \frac{l_1 \theta_1^2}{l_2} \sin(\theta_1 - \theta_2) - \frac{g}{l_2} \sin(\theta_2)$$

It follows that:

$$\begin{pmatrix} 1 & \alpha_1 \\ \alpha_2 & 1 \end{pmatrix} \begin{pmatrix} \ddot{\theta}_1 \\ \ddot{\theta}_2 \end{pmatrix} = \begin{pmatrix} f_1 \\ f_2 \end{pmatrix}$$

Multiplying the inverse of the first matrix gives:

$$\begin{pmatrix} \ddot{\theta}_1 \\ \ddot{\theta}_2 \end{pmatrix} = \frac{1}{1 - \alpha_1 \alpha_2} \begin{pmatrix} 1 & -\alpha_1 \\ -\alpha_2 & 1 \end{pmatrix} \begin{pmatrix} f_1 \\ f_2 \end{pmatrix} \therefore \ddot{\theta}_1 = \frac{f_1 - \alpha_1 f_2}{1 - \alpha_1 \alpha_2} \quad \text{and} \quad \ddot{\theta}_2 = \frac{f_2 - \alpha_2 f_1}{1 - \alpha_1 \alpha_2}$$

This can be used with the fourth-order Runge-Kutta method (RK4) to generate numerical solutions.